

Ordinary paper converted into flexible electronic keypad

Purdue University engineers have developed an amazing printing technique that renders any paper or cardboard packaging into a keyboard, keypad, or other easy-to-use human-machine interfaces. It was developed using Teflon, a chemical coating that keeps food from sticking to pots and pans, and makes paper waterproof, dustproof, and oilproof.

The electronic circuit was created by including layers that contained tiny nickel particles, which acted as wires for conducting electricity. A tiny Bluetooth chip was added, which allowed the paper to communicate with a computer. In the new keypad, the pressure of a finger-tap acts as a power source by sending electricity along the printed wires. This innovation is suitable for low-power devices such as wearables.

Purdue University engineers have developed a simple printing process that renders any paper or cardboard packaging into a keyboard, keypad, or other easy-to-use human-machine interfaces (Credit: www.purdue.edu)



New transistor design disguises key computer chip from hackers



A hacker can reproduce a circuit on a chip by discovering what key transistors are doing in a circuit. Now Purdue University engineers have demonstrated a way to disguise which transistor is which by building them from material called black phosphorus so that, when a voltage toggles the transistors' type, they appear exactly the same to a hacker.

Current camouflaging techniques require more transistors in order to hide what's going on in the circuit. But hiding the transistor type using black phosphorus—a material as thin as an atom—requires fewer transistors, taking up less space and power in addition to creating a better disguise. This built-in security measure would prevent hackers from getting enough information about the circuit to reverse engineer it.

The four transistors on this chip were built out of a 2D material that disguises them from hackers (Credit: Purdue University photo/John Underwood)

New magnetic spray converts objects into micro robots for biomedical applications

An easy way to make milli-bots is by coating objects with a glue-like magnetic spray known as M-spray. This development was made in a joint research led by a scientist from City University of Hong Kong (CityU). Driven by the magnetic field, the coated objects were able to crawl, walk, or roll on different surfaces.

Composed of polyvinyl alcohol (PVA), gluten, and iron particles, M-spray can adhere to the rough and smooth surfaces of 1D, 2D, or 3D objects firmly. The film it forms on the surface is just about 0.1 to 0.25mm thick, which is



The team successfully converted cotton thread, origami, polydimethylsiloxane (PDMS) film and plastic pipe into soft reptile robot, multi-foot robot, walking robot, and rolling robot, respectively (Credit: City University of Hong Kong)